

Terrain-Conditioned Cross-Attention for Cross-Region Avalanche Debris Mapping with Sentinel-1 SAR

Akshay Sharma^{1,*}, Lalji Prasad¹

¹Department of Computer Science and Engineering, SAGE University, Indore 452020, Madhya Pradesh, India

*Corresponding author: Akshay Sharma, akssha74@gmail.com

Article type: Research Article

Target journal: Journal of the Indian Society of Remote Sensing

Running title: Terrain-Conditioned Avalanche Mapping

Main-text word count: 2,045 words

Excludes abstract, tables, figure legends, and references

Figures: 2 Tables: 3 Online resources: 1

Highlights

- The model is tested across four mountain systems and seven avalanche events.
- Three training runs show parity with strong alternatives and a gain only over simple differencing.
- Terrain inputs help in some regions but do not improve accuracy uniformly.
- The slope prior penalizes some pixels that contain genuine avalanche debris.
- The road-corridor test succeeds in the Pamirs but misses the only corridor avalanche in northern Norway.

Author Contributions

Akshay Sharma: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Visualization, Writing—Original Draft.

Lalji Prasad: Supervision, Methodology, Resources, Validation, Writing—Review and Editing.

Acknowledgements

The authors have no acknowledgements to declare.

Funding

No external grant funding was received for this study.

Competing Interests

The authors declare no competing financial or non-financial interests.

Ethics Approval and Consent

Not applicable; this study uses public remote-sensing data and involves no human participants or animals.

Consent for Publication

Not applicable.

Data and Code Availability

The AvalCD benchmark is available at <https://doi.org/10.5281/zenodo.15863589>. Code, checkpoints, the JISRS manuscript source, and verification artifacts are archived at <https://github.com/akssha74/TopoRadar-Net-AvalCD/releases/tag/v1.0.24>.

Author Agreement

Both authors approved the manuscript and agree to its submission. The manuscript has not been published and is not under consideration elsewhere.